

YGNACIO MARTINEZ SANCHEZ

ygnaciomarts.com | Torreón, Coahuila, México

CAREER SUMMARY

Software Engineer Intern with experience building scalable enterprise applications across cloud and frontend environments. Skilled in Node.js, React, and AWS, with a strong focus on system design, automation, and performance. Proven ability to deliver production-ready solutions and optimize developer workflows in large-scale manufacturing environments.

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER INTERN | JOHN DEERE

Aug 2024 – Present

- **Cloud Architecture & Systems Design:**
Designed and deployed a serverless file-handling system using AWS Lambda and S3 Presigned URLs to support document uploads (e.g., training certifications) in a globally used production application for operator certification, overcoming API Gateway payload limits and enabling reliable handling of larger files across manufacturing workflows.
- **Automated Release Engineering:**
Standardized release processes across 10+ repositories by implementing Semantic Release, enabling automated versioning, consistent image tagging, and simplified rollbacks—eliminating manual dependency on commit SHAs and improving deployment reliability.
- **Backend Integration & System Migration:**
Led the migration of critical service endpoints from legacy Java environments to a modern Node.js architecture as part of a large-scale enterprise manufacturing platform initiative, enabling scalable backend services for high-usage internal applications.
- **Enterprise UI Architecture:**
Led and coordinated the migration and full UI redesign implementation of a large-scale enterprise application using a modern component library (Material UI-based), ensuring consistency across the system and successfully delivering the upgrade with no major production incidents.
- **Scalable UI Systems:**
Engineered reusable UI components using Material UI (MUI), adopted by ~5 teams and growing, reducing UI inconsistencies and eliminating duplicated implementations across enterprise applications.
- **Infrastructure & Reliability:**
Diagnosed and resolved container-level issues including crashes, memory constraints, and deployment failures in containerized (CaaS) environments, improving system stability and reducing friction in development and deployment workflows.

EDUCATION

BACHELOR'S DEGREE | Computer Systems Engineering | Instituto Tecnológico de La Laguna | 2020 – Present
Expected Graduation: Jul 2026

HIGH SCHOOL DIPLOMA | Centro de Bachillerato Tecnológico Agropecuario No. 172 | 2017 – 2020
Computer Technical Degree

WORKSHOPS/ACHIEVEMENTS

- International Collegiate Programming Contest (ICPC) | Instituto Tecnológico de La Laguna | 2022, 2023, 2024
- Internal Hackathon – Cloud-based Assistant (Microsoft Stack) | John Deere | 2024

TECHNICAL SKILLS

- Cloud & DevOps Architecture: Containers as a Service (CaaS), CI/CD Strategy & Pipelines, AWS (Lambda, S3, API Gateway, IAM), Microsoft Azure, Docker, GitHub Actions, Monitoring & Observability, Release Automation
- Backend: Node.js, Express.js, Java, RESTful API Design, Authentication & Authorization (Okta), Data Migration
- Data & Persistence: Relational Database Design & Optimization (SQL, PostgreSQL, MySQL)
- Frontend: React, TypeScript, Material UI (MUI), Performance Optimization, Scalable UI Patterns
- Design & UX: UI/UX Principles, Wireframing, Prototyping, Adobe Photoshop

LANGUAGES

- Spanish (Native)
- English (B2)